Lecture 10 (*March 10*)

Reading: Section 3.5
Viewing: Lecture 10

To a $m \times n$ matrix A, we associated the following subspaces:

$$\text{the column space } C(A) \subset \mathbb{R}^m$$
$$\text{the nullspace } N(A) \subset \mathbb{R}^n$$

We will now consider two more subspaces corresponding to the matrix A:

$$\text{the row space } C(A^T) \subset \mathbb{R}^n$$
$$\text{the left nullspace } N(A^T) \subset \mathbb{R}^m$$

Collectively, the four subspaces above are called the **four fundamental subspaces** of A. The row space is just the subspace spanned by the rows of the matrix A (which are the same things as the columns of the transpose matrix A^T). Meanwhile, the left nullspace consists of $m \times 1$ vectors such that:

$$A^T \begin{bmatrix} v_1 \\ \vdots \\ v_m \end{bmatrix} = 0$$

If you transpose the equation above, then you obtain (using formula (43)):

$$\begin{bmatrix} v_1 & \dots & v_m \end{bmatrix} A = 0$$

hence the left nullspace of A consists of $1 \times m$ vectors (in other words, row vectors) $\boldsymbol{v}$ such that:

$$\boxed{\boldsymbol{v}A = 0} \tag{84}$$

This equation explains why we call it the "left" nullspace.

Fact 8. *If r is the rank of the matrix A (number of pivots), then the four subspaces have dimensions:*

$$\boxed{\dim C(A) = r} \qquad \boxed{\dim C(A^T) = r}$$

$$\boxed{\dim N(A) = n - r} \qquad \boxed{\dim N(A^T) = m - r}$$

Let's try and justify the fact above. The fact that the column space has dimension equal to r is just a consequence of the fact that the pivot columns form a basis of $C(A)$. The fact that the nullspace has dimension $n - r$ holds because a choice of basis vectors of $N(A)$ is given by setting all the free variables equal to 0 except for one of them; the number of such basis vectors is the same as the number of free variables, namely n minus the number of pivot variables, i.e. $n-r$.

Let's justify the fact that the row space of A also has dimension r. To see this, let us bring the matrix in reduced row echelon form R (row operations do not change the row space). Of the m rows of R, exactly r of them will be non-zero (the rows which have pivots) and $m - r$ of them will be zero (the rows which do not have pivots). Therefore, the dimension of the row space is exactly r, since pivot rows are linearly independent. This implies that $\dim C(A^T) = r$, which also implies that

$\dim N(A^T) = m - r$, just by repeating the argument in the previous paragraph for A replaced by A^T.

The reason why we introduced the row space and the left nullspace is that they provide complements for the nullspace and column space, respectively. To understand what this means, suppose you had two vector subspaces:

$$V, W \subset \mathbb{R}^n$$

of dimensions k and $n-k$, respectively. If V and W are sufficiently general, then their intersection only consists of the zero vector, in which case we call V and W **complementary** subspaces. This notion only applies when the dimensions of V and W add up to n, the dimension of ambient space.

Example 3. *Two different lines passing through the origin are complementary subspaces of $\mathbb{R}^2$, and any plane and line which only intersect at the origin are complementary subspaces of $\mathbb{R}^3$.*

However, two planes passing through the origin can never be complementary subspaces in $\mathbb{R}^3$, because their intersection always contains a line (moreover, the dimensions add up to $2+2=4 \neq 3$).

Here is a fact we will understand more closely in upcoming weeks:

The row space and nullspace of A are complementary subspaces of $\mathbb{R}^n$

The column space and left nullspace of A are complementary subspaces of $\mathbb{R}^m$

for any $m \times n$ matrix A. But practically, how do we compute the four fundamental subspaces of a matrix? For example, how do we find a basis of each of them? For the column space and nullspace of A, we saw that the key role was played by its reduced row echelon form R:

- the column space $C(A)$ is spanned by the pivot columns of A (i.e. those columns where the reduced row echelon form R has its pivots)

- the nullspace $N(A)$ is spanned by the vectors $\boldsymbol{v} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$ where all the free variables except for one take the value 0, and the pivot variables take values prescribed by $R\boldsymbol{v} = 0$

So how do we compute the spaces $C(A^T)$ and $N(A^T)$? The direct way is to just put A^T in reduced row echelon form, and apply the previous two bullets. But quite often, the problem naturally deals with the reduced row echelon form of the matrix A itself:

$$A \overset{\text{RREF}}{\rightsquigarrow} R \tag{85}$$

By analogy with (62), we have:

$$\boxed{C(A^T) = C(R^T)} \tag{86}$$

which happens because row operations do not change the row space of a matrix. Therefore:

a basis for $C(A^T)$ is given by the pivot rows of the matrix R

What about the left nullspace $N(A^T)$ in terms of the reduced row echelon form (85)? As we have seen a few lectures ago, Gauss-Jordan elimination involves multiplying the matrix A on the left:

$$KA = R$$

by an invertible matrix K. By (84), a row vector $\boldsymbol{v}$ lies in $N(A^T)$ precisely means:

$$0 = \boldsymbol{v}A = \underbrace{\boldsymbol{v}K^{-1}}_{\text{call this } \boldsymbol{w}} R \tag{87}$$

where $\boldsymbol{v}$ is an $1 \times m$ vector. The reduced row echelon matrix R will have the form, say:

$$\begin{bmatrix} \boxed{1} & * & * & 0 & * & * & 0 \\ 0 & 0 & 0 & \boxed{1} & * & * & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \boxed{1} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

and $0 = \boldsymbol{w}R$ happens if and only if:

$$\boldsymbol{w} = \begin{bmatrix} 0 & 0 & 0 & * & * \end{bmatrix}$$

(namely the only non-zero entries of $\boldsymbol{w}$ are those corresponding to the zero rows of R). So the vectors $\boldsymbol{v}$ that can satisfy (87) are those of the form:

$$\boldsymbol{v} = \boldsymbol{w}K = \begin{bmatrix} 0 & 0 & 0 & * & * \end{bmatrix} K$$

We conclude that:

a basis for $N(A^T)$ is given by the rows of the square matrix K which correspond to the zero rows of the reduced row echelon form matrix R